

The Infinite-Bagged RSM Remainder R_k : What Holds Exactly, and What Needs Assumptions

June 19, 2026

Abstract

This note separates, as cleanly as possible, the results on the infinite-bagged RSM excess risk $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$ into two tiers: (i) what is *exact* — true under the common-loading structure with no further assumption, no approximation, and no asymptotics; and (ii) what *rate* statements ($R_k^{\text{bag},\infty} \asymp z_k^2$, hence the cube-root rule $k^* \asymp T^{1/3}$) require, namely two explicit, checkable, and minimal assumptions. The exact tier gives a computable two-sided bracket and a complete characterization of the optimal k ; the rate tier is governed entirely by which of two assumptions is binding, mapping the effective exponent $\alpha \in [1, 2]$ to the bounded-coherence/heterogeneity regime.

1 Setup and notation

Assume the common-loading covariance structure of the forecast errors,

$$\Sigma_e = q \mathbf{u} \mathbf{u}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad \tau_i = \sigma_i^{-2} > 0,$$

where $q \geq 0$ is the variance of the common (unforecastable) shock and τ_i is the precision of forecaster i 's idiosyncratic error. Write

$$A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \quad \ell_i = \frac{\tau_i}{A_m}, \quad \sum_{i=1}^m \ell_i = 1,$$

and for a subset S of size k , $A_S = \sum_{j \in S} \tau_j$. RSM draws subsets S uniformly at random, fits the constrained (sum-to-one) optimal weights $v_{S,i} = \mathbf{1}\{i \in S\} \tau_i / A_S$ within each, and bags. The infinite-bagging ($G \rightarrow \infty$) weight is $\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}]$, and the object of interest is the population excess risk

$$R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \bar{v}_k' \Sigma_e \bar{v}_k - Q^{\text{opt}}.$$

Two population functionals carry the entire story:

$$V_\delta = \frac{1}{m} \sum_{i=1}^m \left(\frac{\tau_i}{\bar{\tau}} - 1 \right)^2 = \frac{\text{Var}(\tau)}{\bar{\tau}^2} \quad (\text{normalized precision dispersion}), \quad H = \overline{\sigma^2} \bar{\tau} = \left(\frac{1}{m} \sum_i \sigma_i^2 \right) \bar{\tau} \geq 1,$$

the latter being the arithmetic-vs-harmonic gap (Cauchy–Schwarz), with $H = 1$ iff all precisions are equal. Throughout, $z_k = \frac{1}{k} - \frac{1}{m} = \frac{m-k}{km}$.

2 Tier I: what holds exactly (no assumptions)

Every statement in this section is an identity or inequality that holds for an arbitrary positive precision vector $(\tau_1, \dots, \tau_m)$ and every $1 \leq k \leq m$. No balanced-leverage, no distributional assumption, no Taylor expansion, no $T \rightarrow \infty$ or $m \rightarrow \infty$ limit.

Proposition 1 (Oracle weights are leverage scores). *The constrained-optimal full combination is $w_i^* = \tau_i/A_m = \ell_i$, with $Q^{\text{opt}} = q + 1/A_m$, and for any subset $Q(S) = q + 1/A_S$.*

Proof. Since $\boldsymbol{\iota}'w = 1$, $w'q\boldsymbol{\iota}'w = q$ is constant, so minimizing $w'\Sigma_e w$ reduces to minimizing $\sum_i w_i^2 \sigma_i^2$ subject to $\sum_i w_i = 1$; the Lagrange solution is $w_i \propto \tau_i$. Substituting gives $Q^{\text{opt}} = q + \sum_i \ell_i^2 / \tau_i = q + 1/A_m$. The subset claim is the same argument restricted to S . $\square$

Proposition 2 (Exact quadratic-form identity). *Because w^* is the constrained minimizer, the cross term vanishes and*

$$R_k^{\text{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \frac{1}{A_m} \sum_{i=1}^m \ell_i \left(\frac{\bar{v}_{k,i}}{\ell_i} - 1 \right)^2 = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i - \frac{1}{\bar{\tau}} \right)^2,$$

where $g_i(k) = k \mathbb{E}_S[1/A_S \mid i \in S]$ and $\bar{v}_{k,i} = \tau_i g_i(k)/m$.

Proof. $\Sigma_e w^* = (q + 1/A_m)\boldsymbol{\iota} = Q^{\text{opt}}\boldsymbol{\iota}$, and since $\boldsymbol{\iota}'(\bar{v}_k - w^*) = 0$ (both weight vectors sum to one), $(\bar{v}_k - w^*)' \Sigma_e w^* = 0$. The common factor $q\boldsymbol{\iota}'$ then drops for the same reason, leaving $\sum_i (\bar{v}_{k,i} - \ell_i)^2 / \tau_i$; the two displayed forms are algebraic rewrites using $w_i^* = \ell_i$ and $\tau_i = A_m \ell_i$. $\square$

Proposition 3 (Endpoints: RSM bridges averaging and the oracle). *At $k = 1$, $\bar{v}_1 = \frac{1}{m}\boldsymbol{\iota}$ (simple averaging), so R_1 equals exactly the excess risk of the equal-weight combination. At $k = m$, $\bar{v}_m = w^*$, so $R_m = 0$. Thus k traces, in population, a path from simple averaging to the oracle combination.*

Lemma 1 (Two exact moment identities). *For all k ,*

$$\sum_{i=1}^m \tau_i g_i = m \quad (\text{conservation law}), \quad \sum_{i=1}^m g_i = km \mathbb{E}_S[1/A_S].$$

Proposition 4 (Jensen sandwich). *For all $1 \leq k \leq m$,*

$$0 \leq R_k^{\text{bag},\infty} \leq Q_k^{\text{sub}} - Q^{\text{opt}} = \mathbb{E}_S[1/A_S] - \frac{1}{A_m}.$$

Proof. Lower bound: w^* is the global sum-to-one minimizer and $\bar{v}_k$ is feasible. Upper bound: $Q_k^{\text{bag},\infty} = (\mathbb{E}_S v_S)' \Sigma_e (\mathbb{E}_S v_S) \leq \mathbb{E}_S[v_S' \Sigma_e v_S] = Q_k^{\text{sub}}$ by Jensen ($\Sigma_e \succeq 0$). Subtract Q^{opt} and use $Q(S) = q + 1/A_S$, $Q^{\text{opt}} = q + 1/A_m$. $\square$

Proposition 5 (Exact two-sided bracket). *For all $1 \leq k \leq m$,*

$$\frac{(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}} \leq R_k^{\text{bag},\infty} \leq \mathbb{E}_S[1/A_S] - \frac{1}{A_m}.$$

Both ends are computable exactly from $\{\tau_i\}$ and the scalar $\mathbb{E}_S[1/A_S]$. The denominator $\sum_i \sigma_i^2 - m/\bar{\tau} \geq 0$ by Cauchy–Schwarz, with equality iff all τ_i are equal (in which case $R_k^{\text{bag},\infty} = 0$). The lower bound is exact whenever the precision profile takes at most two distinct values, and a convergent hierarchy of exact lower bounds follows by appending higher moment identities to Lemma 1.

Proof. The upper bound is Proposition 4. For the lower bound, minimize $\sum_i \tau_i g_i^2$ over all vectors satisfying the two identities of Lemma 1. In the inner product $\langle x, y \rangle_\tau = \sum_i \tau_i x_i y_i$ these read $\langle g, \mathbf{1} \rangle_\tau = m$ and $\langle g, c \rangle_\tau = km \mathbb{E}_S[1/A_S]$ with $c_i = 1/\tau_i$. With $\|\mathbf{1}\|_\tau^2 = A_m$, $\langle c, \mathbf{1} \rangle_\tau = m$, and $\|c\|_\tau^2 = \sum_i 1/\tau_i$,

$$\min \sum_i \tau_i g_i^2 = \frac{m}{\bar{\tau}} + \frac{m^2(k\mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i 1/\tau_i - m/\bar{\tau}}.$$

Substituting into Proposition 2 and using $\frac{1}{m^2} \cdot \frac{m}{\bar{\tau}} = 1/A_m$, the oracle term cancels. $\square$

Corollary 1 (The unconditional open bracket). *Expanding both ends of Proposition 5 to leading order, using the exact moments $\mathbb{E}_S[A_S] = k\bar{\tau}$, $\text{Var}_S(A_S) = k\frac{m-k}{m(m-1)}V$ with $V = \sum_i (\tau_i - \bar{\tau})^2 = m\bar{\tau}^2 V_\delta$, and $\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$,*

$$\underbrace{\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2}_{\text{floor}} \lesssim R_k^{\text{bag}, \infty} \lesssim \underbrace{\frac{1}{\bar{\tau}} z_k}_{\text{ceiling}}.$$

Remark 1 (The two ends answer to different things). *Since $0 < z_k < 1$ we have $z_k^2 < z_k$, so the bracket has a gap between exponents: unconditionally one can only conclude $R_k^{\text{bag}, \infty} \sim z_k^\alpha$ with $\alpha \in [1, 2]$, hence $k^* \in [\Theta(T^{1/3}), \Theta(T^{1/2})]$. Crucially the gap is asymmetric:*

- the **floor** z_k^2 is produced by the dispersion V_δ (numerator of the Gram bound) and is present whenever the forecasters are heterogeneous — coherence plays no role in the exponent;
- the **ceiling** z_k is the Jensen/average-subset bound, which is intrinsically first order because it cannot see that bagging cancels the leading term.

The effective exponent is therefore read off the ceiling: the floor is always z_k^2 , and where α actually lands is decided by how far the ceiling can be pulled down. This is the entire role of Tier II: Assumption A moves the ceiling, Assumption B sets the floor.

Proposition 6 (Strict monotonicity). *If each $g_i(k)$ is monotone in k , then $R_k^{\text{bag}, \infty}$ is strictly decreasing in k for any non-degenerate precision profile:*

$$\frac{dR_k^{\text{bag}, \infty}}{dk} = \frac{2}{m^2} \sum_i \tau_i \left(g_i - \frac{1}{\bar{\tau}}\right) \dot{g}_i < 0.$$

Proof. From Proposition 2 and $\sum_i \tau_i \dot{g}_i = 0$ (differentiating the conservation law), $\frac{dR_k^{\text{bag}, \infty}}{dk} = \frac{2}{m^2} \sum_i \tau_i (g_i - 1/\bar{\tau}) \dot{g}_i$. Each g_i runs from $g_i(1) = 1/\tau_i$ to $g_i(m) = 1/\bar{\tau}$; under monotonicity, a high-precision coordinate ($\tau_i > \bar{\tau}$) has $g_i \leq 1/\bar{\tau}$ throughout with $\dot{g}_i \geq 0$, and a low-precision one the reverse, so every product is ≤ 0 , strictly for at least one i unless all τ_i are equal. $\square$

Remark 2. *The monotonicity of each g_i is the one lemma in Tier I that requires its own proof; the unconditional bound $R_k^{\text{bag}, \infty} \leq R_1$ (Jensen on the harmonic mean) survives without it. Proposition 6 has a sharp consequence: in population there is no bias–variance trade-off in k — larger k is always weakly better. An interior optimum exists only once estimation error enters through the finite- T inflation factor below.*

Proposition 7 (Exact optimal- k condition). *Let $A = \sigma_\varepsilon^2 + Q^{\text{opt}}$ and let the idealized infinite-bagged loss be $L_k = (A + R_k^{\text{bag},\infty}) \frac{T-1}{T-k}$. The discrete optimum k^* satisfies*

$$(T - k^*)(R_{k^*} - R_{k^*+1}) = A + R_{k^*}.$$

Proposition 8 (RSM dominates CSR). *For every k , $Q_k^{\text{RSM}} \leq Q_k^{\text{CSR}}$: within each subset the constrained-OLS weight minimizes $w' \Sigma_e w$ over the equal-weight choice $1/k$; take expectations over S .*

3 Tier II: the assumptions needed for a rate

The unconditional bracket of Corollary 1 is already two-sided, but its two exponents differ: floor z_k^2 , ceiling z_k . By Remark 1 the floor is fixed by dispersion alone, so the only thing standing between us and a genuine rate $R_k^{\text{bag},\infty} \asymp z_k^2$ is closing the *ceiling* from z_k down to z_k^2 . This requires exactly two assumptions, and they are not interchangeable: Assumption 1 sharpens the ceiling, Assumption 2 guarantees the floor is non-degenerate.

Assumption 1 (Bounded coherence). *There exist constants $0 < c \leq C < \infty$, independent of m , such that $c \leq \tau_i/\bar{\tau} \leq C$ for all i .*

Role. The upper part (C) is “no dominant forecaster”: $\kappa = \tau_{\max}/\bar{\tau} \leq C$. The lower part (c) is “no negligible forecaster”: it forces $A_S \geq ck\bar{\tau}$, keeping $1/A_S$ and all its reciprocal moments bounded. This is what makes the expansion $\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$ rigorous, with an explicit, bounded remainder, rather than a heuristic.

Assumption 2 (Non-degenerate dispersion). *There exists $v > 0$, independent of m , such that $V_\delta \geq v$.*

Role. Genuine heterogeneity. Without it $R_k^{\text{bag},\infty} \rightarrow 0$ and there is nothing to bound (simple averaging is near-optimal). The upper bound $V_\delta \leq (C-1)^2$ and $H \leq C/c$ are already implied by Assumption 1.

Theorem 1 (Two-sided z_k^2 bound). *Under Assumptions 1 and 2 there are constants $0 < c_1 \leq c_2 < \infty$, depending only on c, C, v , such that for all $2 \leq k < m$,*

$$\frac{c_1}{m\bar{\tau}} z_k^2 \leq R_k^{\text{bag},\infty} \leq \frac{c_2}{m\bar{\tau}} z_k^2.$$

Consequently the effective exponent is $\alpha = 2$ exactly, and the optimal subset size satisfies $k^ \asymp T^{1/3}$.*

Proof sketch. *Upper.* Under Assumption 1, the reciprocal expansion of $\mathbb{E}_R[1/L_{i,R}]$ has a remainder controlled by $A_S \geq ck\bar{\tau}$; combined with the SRSWOR variance bound $\text{Var}_R(L_{i,R}) \leq \frac{(k-1)(m-k)}{m(m-1)(m-2)} V_\delta$ this gives $|\bar{v}_{k,i}/\ell_i - 1| \leq K_c z_k (|h_i| + V_\delta)$ with $h_i = \tau_i/\bar{\tau} - 1$. Plugging into Proposition 2 and using $\ell_i \leq C/m$, $\sum_i \ell_i h_i^2 \leq CV_\delta$, yields $R_k^{\text{bag},\infty} \leq \frac{K_{c,C}}{m\bar{\tau}} (V_\delta + V_\delta^2) z_k^2$. *Lower.* In Proposition 5, $k\mathbb{E}_S[1/A_S] - 1/\bar{\tau} \asymp V_\delta z_k$ and the denominator $\sum_i \sigma_i^2 - m/\bar{\tau} = m\bar{\tau}^{-1}(H-1)$ is bounded by Assumption 1; with $V_\delta \geq v$ this gives $R_k^{\text{bag},\infty} \geq \frac{v^2}{m(H-1)\bar{\tau}} z_k^2$. The rate for k^* follows by inserting $R_k^{\text{bag},\infty} \asymp C_R z_k^2$ (with $z_k \approx 1/k$ for $k \ll m$) into Proposition 7: $Ak^3 + 3C_R k - 2C_R T = 0$, so $k^* \approx (2C_R T/A)^{1/3}$. $\square$

Remark 3 (How each assumption acts, and two caveats). *Relative to Corollary 1, the theorem changes only the ceiling: Assumption 1 replaces the Jensen z_k bound by a z_k^2 bound, collapsing the bracket onto the floor that Assumption 2 already pins at z_k^2 . Two caveats keep this honest.*

1. The floor stays valid but loosens under high coherence. If a dominant forecaster does appear, the true $R_k^{\text{bag},\infty}$ rises toward the z_k ceiling ($\alpha \rightarrow 1$), so the z_k^2 floor remains a correct lower bound but is no longer tight. A floor never certifies $\alpha = 2$; only the ceiling coming down does.
2. The floor's constant is uniform in m only under Assumption 1. The lower constant is $\propto V_\delta^2 / (m(H-1)\bar{\tau})$, and H is bounded uniformly in m only under 1 (a near-negligible forecaster makes H blow up and the floor sag). For a fixed profile this is irrelevant — H is a finite number and the floor is a fixed constant times z_k^2 — so the z_k^2 exponent on the floor is owed to 2 alone; only its uniform constant borrows from 1.

4 The explicit leading constant

Theorem 1 shows $R_k^{\text{bag},\infty} \asymp z_k^2$ with constants c_1, c_2 left abstract. Under the same assumptions one can compute the leading constant in closed form from the first three moments of the precision profile. The result instantiates $c_1 = c_2 = D_\tau$ (up to the $(m/(m-1))^2$ factor) *inside* the bracket; it is a leading-order approximation, not a new bound.

Recall $h_i = \tau_i/\bar{\tau} - 1$ (so $\sum_i h_i = 0$ and $V_\delta = \frac{1}{m} \sum_i h_i^2$), and define the third moment and the *weighted heterogeneity constant*

$$M_3 = \frac{1}{m} \sum_{i=1}^m h_i^3, \quad C_\tau = \frac{1}{m} \sum_{i=1}^m (1+h_i)(h_i - V_\delta)^2 = V_\delta + M_3 - V_\delta^2.$$

Since $1+h_i = \tau_i/\bar{\tau} = m\ell_i > 0$, we have $C_\tau = \sum_i \ell_i (h_i - V_\delta)^2 \geq 0$: it is exactly the variance of h under the leverage (precision-weighted) distribution $\ell_i = \tau_i/A_m$, linking it to the coherence picture. If heterogeneity is mild, M_3, V_δ^2 are higher order and $C_\tau \approx V_\delta$.

Lemma 2 (Conservation-consistent first-order expansion). *Under Assumption 1, to first order in z_k ,*

$$g_i(k) = \frac{1}{\bar{\tau}} \left[1 + \frac{m}{m-1} z_k (V_\delta - h_i) \right] + o(z_k).$$

Derivation. Conditioning on $i \in S$, $\mathbb{E}_S[A_S \mid i \in S] = k\bar{\tau} \left[1 + \frac{m}{m-1} z_k h_i \right]$, so the delta method gives $g_i \approx \frac{1}{\bar{\tau}} \left[1 - \frac{m}{m-1} z_k h_i \right]$. This violates the exact identity $\sum_i \tau_i g_i = m$ (Lemma 1) by $-\frac{m}{m-1} z_k m V_\delta$. The discrepancy is removed by recentering h_i at its *precision-weighted* mean, which is exactly V_δ : $\frac{\sum_i \tau_i h_i}{\sum_i \tau_i} = \frac{1}{m} \sum_i (h_i + h_i^2) = V_\delta$. Replacing h_i by $h_i - V_\delta$ restores $\sum_i \tau_i g_i = m$ identically and yields the stated form; the second-order reciprocal term contributes at $o(z_k)$ under Assumption 1. $\square$

Proposition 9 (Closed-form leading constant). *Under Assumptions 1–2, with $B = 1/\bar{\tau}$,*

$$R_k^{\text{bag},\infty} \approx D_\tau z_k^2, \quad D_\tau = \frac{B}{m} \left(\frac{m}{m-1} \right)^2 C_\tau,$$

equivalently $Q_k^{\text{bag},\infty} \approx Q^{\text{opt}} + D_\tau z_k^2$ with $Q^{\text{opt}} = q + B/m$. In the mild-heterogeneity limit $C_\tau \rightarrow V_\delta$, so $D_\tau \rightarrow V_\delta/(m\bar{\tau})$, recovering the Hájek/ $\alpha = 2$ leading term.

Proof. Substitute Lemma 2 into the exact identity $R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i (g_i - \frac{1}{\bar{\tau}})^2$. With $g_i - \frac{1}{\bar{\tau}} = -\frac{1}{\bar{\tau}} \frac{m}{m-1} z_k (h_i - V_\delta)$,

$$R_k^{\text{bag},\infty} \approx \frac{1}{m^2} \left(\frac{m}{m-1} \right)^2 \frac{z_k^2}{\bar{\tau}^2} \sum_i \tau_i (h_i - V_\delta)^2 = \frac{1}{m\bar{\tau}} \left(\frac{m}{m-1} \right)^2 C_\tau z_k^2,$$

using $\sum_i \tau_i (h_i - V_\delta)^2 = \bar{\tau} \sum_i (1+h_i)(h_i - V_\delta)^2 = \bar{\tau} m C_\tau$. $\square$

Corollary 2 (Explicit cube-root rule). *With $A = \sigma_\varepsilon^2 + Q^{\text{opt}}$ and $L_k = (A + D_\tau z_k^2)^{\frac{T-1}{T-k}}$, the exact first-order condition is $A + D_\tau z_k^2 = \frac{2D_\tau z_k(T-k)}{k^2}$, which for large T (and $z_k \approx 1/k$) gives*

$$k^* \approx \left(\frac{2D_\tau T}{A} \right)^{1/3}.$$

This is a directly computable plug-in: estimate $\bar{\tau}$, V_δ , M_3 , form D_τ , and minimize L_k over integer k .

Remark 4 (Status: approximation, not bound). *Proposition 9 is the leading-order term, obtained from a first-order expansion of $1/A_S$; it is not a two-sided bound and does not supersede Proposition 5. Its role is to supply the explicit value of the constant inside the closed bracket of Theorem 1. Three qualifications: (i) the expansion of $1/A_S$ requires A_S bounded away from zero, i.e. Assumption 1; (ii) it is a single power of z_k , accurate for mild-to-moderate heterogeneity — for strongly heterogeneous (factor-type) profiles the curvature at small k is governed by M_3 and higher moments, so a single leading term understates the small- k decay; (iii) when the precisions τ_i are estimated, D_τ inherits the sampling noise in V_δ and especially M_3 . Within these limits it is the correct and useful closed form for the z_k^2 constant, consistent in the mild limit with the Hájek term and with the $T^{1/3}$ scaling observed in the Monte Carlo.*

5 Convergence along the path

Proposition 3 says $\bar{v}_k$ is a *path* in weight space from $w^{\text{ave}} = \frac{1}{m}\mathbf{1}$ (at $k = 1$) to w^* (at $k = m$), so $R_k^{\text{bag},\infty} = \|\bar{v}_k - w^*\|_\tau^2$ is the squared *distance still to travel*, where $\langle a, b \rangle_\tau = \sum_i a_i b_i / \tau_i$. Bounding $R_k^{\text{bag},\infty}$ is then a question about how the weights converge, and heterogeneity governs the shape of that convergence. Write the per-coordinate fractional progress

$$\pi_i(k) = \frac{\bar{v}_{k,i} - 1/m}{\ell_i - 1/m} \in [0, 1], \quad \pi_i(1) = 0, \quad \pi_i(m) = 1.$$

Proposition 10 (Homogeneous convergence $\Rightarrow$ straight line). *If the coordinates progress uniformly, $\pi_i(k) \equiv \pi(k)$, then the path is the line $w^{\text{ave}} \rightarrow w^*$, the deviation $\delta_k = \bar{v}_k - w^*$ stays parallel (in the τ -metric) to $\eta = w^{\text{ave}} - w^*$, and*

$$R_k^{\text{bag},\infty} = (1 - \pi(k))^2 R_1, \quad R_1 = \frac{H - 1}{m\bar{\tau}}.$$

In this case the Gram/projection lower bound of Proposition 5 holds with equality.

This recovers the tightness condition of Proposition 5 (equality for ≤ 2 distinct precision types) in path language: uniform convergence $\Leftrightarrow \delta_k \parallel \eta \Leftrightarrow$ the projection bound is exact.

Remark 5 (Heterogeneity bends the path). *With genuine heterogeneity the coordinates do not move in lockstep. From Lemma 2, to leading order*

$$\bar{v}_{k,i} - w_i^* \approx z_k u_i, \quad u_i \propto \tau_i(h_i - V_\delta) = \bar{\tau}(1 + h_i)(h_i - V_\delta),$$

so convergence is self-similar in z_k but along the direction u , whereas the projection of Proposition 5 uses the single direction $\eta_i = \frac{1}{m} - \ell_i = -h_i/m$. The two coincide only in the homogeneous/ ≤ 2 -type limit; in general the angle between u and η is the heterogeneity. This is exactly why the single-direction bound is loose:

- projecting onto η keeps only u 's η -component, giving the floor constant $V_\delta^2/(H-1)$ (Cor. 1);
- using the true direction u keeps all of it, giving back $R_k^{\text{bag},\infty} \approx D_\tau z_k^2$ with $C_\tau = \frac{1}{m} \sum_i (1 + h_i)(h_i - V_\delta)^2$ (Prop. 9).

The two agree in the mild limit, where $H-1 \rightarrow V_\delta$ and $C_\tau \rightarrow V_\delta$.

Remark 6 (The exponent is the slowest coordinate's convergence rate). Since $R_k^{\text{bag},\infty} = \sum_i \delta_{k,i}^2 / \tau_i$, the exponent α is set by which weight converges slowest.

- Bounded coherence (heterogeneous, no dominant forecaster): every coordinate converges at the common rate $\delta_{k,i} \sim z_k$, so $R_k^{\text{bag},\infty} \sim z_k^2$ and $\alpha = 2$.
- Dominant forecaster (high coherence): a single coordinate i^* with huge τ_{i^*} is heavily under-weighted at equal weights ($1/m \ll \ell_{i^*}$) and converges slowly, retaining a persistent η -aligned component; its term dominates and $R_k^{\text{bag},\infty} \sim z_k$, so $\alpha \rightarrow 1$.

Thus the bagging exponent is governed by the convergence rate of the slowest-converging weight — the convergence-theoretic face of the floor/ceiling bracket.

Usable form. The path view gives the anchored distance-remaining model

$$R_k^{\text{bag},\infty} = R_1 \phi(z_k)^2, \quad \phi(z_1) = 1, \quad \phi(0) = 0, \quad \phi(z_k) \approx \sqrt{D_\tau/R_1} z_k \quad (\text{leading order}),$$

with ϕ the normalized distance still to travel and heterogeneity setting its shape through u : homogeneous convergence gives $\phi(z_k) = z_k/z_1$ ($\alpha = 2$), a dominant forecaster flattens ϕ toward $\sqrt{z_k}$ ($\alpha \rightarrow 1$), and minimization returns $k^* \propto T^{1/(\alpha+1)}$ as in Corollary 2. To tighten the *lower bound* rigorously, project onto the two-dimensional span $\{\eta, u\}$ (equivalently append the moment $\sum_i \tau_i^2 g_i$): exact for ≤ 3 precision types, the geometric form of the moment hierarchy of Proposition 5.

6 U-statistic structure and finite-ensemble convergence

Two idealizations sit outside the bracket: k is treated as a growing tuning size, and bagging is taken to the infinite-ensemble limit $G \rightarrow \infty$. The generalized-U-statistic theory of Peng, Coleman & Mentch (2022) closes the first, and the algorithmic-convergence analysis of Lopes, Wu & Lee (2019) the second.

The bagged weight is a generalized U-statistic

The infinite-bagged weight $\bar{v}_{k,i} = \mathbb{E}_S[h_i(S)]$ is a complete generalized U-statistic of degree k on the m forecasters, with kernel $h_i(S) = \tau_i/A_S$; the realized RSM weight $\hat{v}_{k,i} = \frac{1}{G} \sum_{g=1}^G v_{S_g,i}$ over G random subsets is its incomplete version. Write $\zeta_k = \text{Var}_S(h_i)$ (full kernel variance) and $\zeta_1 = \text{Var}(\mathbb{E}[h_i | \text{one forecaster fixed}])$ (the Hájek/first-order variance). The variance ratio $\zeta_k/(k\zeta_1)$ measures asymptotic linearity: it tends to 1 exactly when the Hájek projection dominates.

Remark 7 (Hájek projection is the exact leading term). Because $h_i = \tau_i/A_S$ is a smooth, OLS-type functional of the subset precisions, its variance ratio is bounded (Peng–Coleman–Mentch, Prop. 1–4 and App. B.3 for OLS), so $\bar{v}_{k,i}$ is asymptotically linear and the Hájek projection is the leading term. This is the rigorous justification for treating the z_k^2 expansion of Proposition 9 as the exact leading order (not a heuristic): the small-heterogeneity result and the Hájek first-order projection coincide, fixing $\alpha = 2$ under bounded coherence. The Berry–Esseen bound of Peng–Coleman–Mentch (Thm. 3) quantifies the rate of normality as $\propto n^{-1/2} + \{\frac{k}{n}(\frac{\zeta_k}{k\zeta_1} - 1)\}^{1/2}$, making the roles of k and the sample size explicit.

Finite-ensemble ($G < \infty$) inflation

Under common loading the gap between the realized finite- G risk and the infinite-bagged risk is exact and of order $1/G$.

Proposition 11 (Finite- G risk inflation). *Let $\hat{v}_k = \frac{1}{G} \sum_{g=1}^G v_{S_g}$ with $S_1, \dots, S_G$ i.i.d. uniform size- k subsets, and $\hat{Q}_G = \hat{v}_k' \Sigma_e \hat{v}_k$. Since each v_{S_g} sums to one, so does $\hat{v}_k$, and*

$$\mathbb{E}_G[\hat{Q}_G] = Q_k^{\text{bag}, \infty} + \frac{\Gamma_k}{G}, \quad \Gamma_k = \sum_{i=1}^m \frac{\text{Var}_S(v_{S,i})}{\tau_i} \geq 0.$$

Proof. The draws are i.i.d., so $\mathbb{E}_G[\hat{v}_{k,i}] = \bar{v}_{k,i}$ and $\text{Var}_G(\hat{v}_{k,i}) = \text{Var}_S(v_{S,i})/G$. As in Proposition 2 the common factor and cross terms drop for a sum-to-one vector, giving $\hat{Q}_G = q + \sum_i \hat{v}_{k,i}^2 / \tau_i$; take expectations and use $\mathbb{E}_G[\hat{v}_{k,i}^2] = \bar{v}_{k,i}^2 + \text{Var}_S(v_{S,i})/G$. $\square$

Remark 8 (Reading Γ_k and choosing G). Γ_k is the ensemble (Monte-Carlo) variance — the common-loading analogue of the Peng–Coleman–Mentch incomplete- U -statistic term ζ_k/N with $N = G$, and it matches the $O(1/t)$ algorithmic-convergence rate of Scornet (2016) and Lopes–Wu–Lee. It decreases in k : at $k = 1$, $v_{S,i}$ is Bernoulli($1/m$) with variance $\frac{1}{m}(1 - \frac{1}{m})$, while at $k = m$ it is the deterministic ℓ_i with zero variance. Hence **small k needs more bags**. Practically, require the finite- G inflation Γ_k/G to be small relative to the bagging gain $R_1 - R_k$; or, without computing Γ_k , use the Lopes–Wu–Lee ensemble bootstrap — resample the G subset-weights with replacement to estimate the $(1 - \alpha)$ -quantile $q_{1-\alpha}(G)$ of $\hat{Q}_G - Q_k^{\text{bag}, \infty}$ from a single run — and raise G until $q_{1-\alpha}(G)$ is below tolerance. Lopes–Wu–Lee (§2.2) also note that some coordinates’ weights converge more slowly, distorting them: the same slowest-coordinate effect as Remark 6.

Remark 9 (Two error sources in the endogenous case). When the weights are estimated from T observations there are two independent error sources, and the incomplete- U -statistic variance takes the Peng–Coleman–Mentch form

$$\approx \underbrace{\frac{C_1}{T}}_{\text{estimation (data)}} + \underbrace{\frac{\Gamma_k}{G}}_{\text{ensemble}}.$$

The variance ratio $\zeta_k/(k\zeta_1)$ governs which dominates; the two are comparable when $G \sim T\Gamma_k/C_1$, so $G \sim T$ suffices and $G \gg T$ is unnecessary. The data term is the estimation-error/Kan–Zhou contribution that drives the gap between the exogenous and endogenous optimal k .

7 Minimality: each failure mode collapses one side

The two assumptions are tight; because each governs a different end of the bracket, dropping either one degrades the bound in a specific, interpretable way. The following table reads off the floor and ceiling under each combination.

Assumptions	Floor	Ceiling	α	k^*
1 and 2	z_k^2	z_k^2	2	$\Theta(T^{1/3})$
2 only (no 1)	z_k^2	z_k	$[1, 2]$	$[\Theta(T^{1/3}), \Theta(T^{1/2})]$
1 only (no 2)	0	z_k^2	$[0, 2]$	up to $\Theta(T^{1/3})$
neither	0	z_k	$[0, 2]$	—

The middle row is the case discussed throughout: drop bounded coherence and the *floor is unchanged* at z_k^2 (it is owed to dispersion, 2), while the *ceiling springs back* from z_k^2 to the Jensen value z_k — precisely the open bracket of Corollary 1. The economic content of each failure mode:

Assumption dropped	Regime	What survives	Rate
1-lower ($\tau_{\min}/\bar{\tau} \rightarrow 0$)	dominant/negligible forecaster	only Jensen, $R_k^{\text{bag},\infty} \lesssim z_k$	$k^* \asymp T^{1/2}$
2-lower ($V_\delta \rightarrow 0$)	homogeneous forecasters	$R_k^{\text{bag},\infty} \rightarrow 0$	no tuning; averaging optimal

Thus the cube root is precisely the *bounded-coherence, heterogeneous* regime; the square root and the degenerate case are its two boundaries. The effective exponent $\alpha \in [1, 2]$ is therefore not a fitted quantity but a map of which assumption is binding — and, by Remark 1, it is the ceiling that moves.

8 Discussion: plausibility, and a fixed- m alternative

Empirical content. The two assumptions are checkable, and they align with the applications. Assumption 2 (heterogeneity) is the “there are gains” condition: it holds in the heterogeneous vintage panels, where RSM delivers real gains, and fails for near-homogeneous professional-forecaster consensus data, where gains are negligible — exactly the dividing line the empirical work traces. Assumption 1 (no dominant forecaster) is the same economic point made in the forecast-combination literature: a single super-precise or strongly negatively correlated forecaster is the case where averaging fails badly and is empirically unlikely; assuming 1 assumes that pathology away.

Uniformity is only for the large- m story. Assumptions 1 and 2 are stated with constants independent of m because that is what a *rate in m* (a stable exponent as the panel grows) requires. If only a fixed panel is of interest, no uniformity is needed: the exact bracket (Proposition 5) already supplies finite, computable two-sided bounds, and the only condition required for a non-trivial statement is the non-degeneracy $V_\delta > 0$. In short:

fixed m : $V_\delta > 0$ plus the exact bracket; large m /rate : uniform Assumptions 1 and 2.

Scope. All of the above governs the population object $R_k^{\text{bag},\infty}$ (infinite bagging). The map from $R_k^{\text{bag},\infty}$ to a realized, finite- G , finite- T mean-squared forecast error still involves the bagging idealization ($G \rightarrow \infty$) and the estimation-error inflation factor $\frac{T-1}{T-k}$, which are modeling choices outside the bracket and should be stated as such.

9 Summary

1. **Exact, no assumptions:** oracle weights are leverage scores (Prop. 1); $R_k^{\text{bag},\infty}$ is an exact quadratic form in the weight deviation (Prop. 2); RSM bridges averaging ($k = 1$) and the oracle ($k = m$) (Prop. 3); a computable two-sided bracket (Prop. 5); strict monotonicity, so no population bias–variance trade-off (Prop. 6); an exact optimal- k condition (Prop. 7); and RSM $\succeq$ CSR (Prop. 8).
2. **Unconditional open bracket:** $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$ (Cor. 1), with the floor z_k^2 set by dispersion and the ceiling z_k the Jensen bound; hence $\alpha \in [1, 2]$, $k^* \in [\Theta(T^{1/3}), \Theta(T^{1/2})]$.

3. **Rate, two assumptions:** *Assumption 1 moves the ceiling* ($z_k \rightarrow z_k^2$); *Assumption 2 sets the floor* (z_k^2). Together the bracket closes: $R_k^{\text{bag},\infty} \asymp z_k^2$ and $k^* \asymp T^{1/3}$ (Thm. 1).
4. **Explicit constant:** under both assumptions the leading constant is closed form, $R_k^{\text{bag},\infty} \approx D_\tau z_k^2$ with $D_\tau = \frac{B}{m}(\frac{m}{m-1})^2 C_\tau$ and $C_\tau = V_\delta + M_3 - V_\delta^2$ the leverage-weighted dispersion of h (Prop. 9); it gives the plug-in $k^* \approx (2D_\tau T/A)^{1/3}$ (Cor. 2) and reduces to the Hájek term $V_\delta/(m\bar{\tau})$ in the mild limit. It is a leading-order approximation inside the bracket, not a new bound (Rem. 4).
5. **Convergence view:** $R_k^{\text{bag},\infty} = \|\bar{v}_k - w^*\|_\tau^2$ is the squared distance-to-go on the averaging→oracle path (Prop. 3). Uniform convergence gives $R_k^{\text{bag},\infty} = (1 - \pi(k))^2 R_1$ and makes the projection bound exact (Prop. 10); heterogeneity bends the convergence direction from η toward $u \propto \tau(h - V_\delta)$ (Rem. 5), and α is the slowest coordinate's convergence rate (Rem. 6).
6. **U-statistic / finite ensemble:** $\bar{v}_{k,i}$ is a generalized U-statistic with kernel τ_i/A_S ; its bounded variance ratio makes the Hájek projection the exact leading term, rigorizing $\alpha = 2$ (Rem. 7). The realized finite- G risk satisfies $\mathbb{E}_G[\hat{Q}_G] = Q_k^{\text{bag},\infty} + \Gamma_k/G$ (Prop. 11), with Γ_k decreasing in k (small k needs more bags); in the endogenous case the variance splits as $C_1/T + \Gamma_k/G$, so $G \sim T$ suffices (Rems. 8–9).
7. **Minimality:** dropping 1 reopens the ceiling to z_k (square-root regime, $\alpha \rightarrow 1$); dropping 2 drops the floor to 0 (degenerate, averaging-optimal). The exponent $\alpha \in [1, 2]$ indexes which assumption binds, and it is the ceiling that moves.